

KAIST
2023 Fall
[MAS456] Statistical Methods with Computer
Homework 3

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Chapter 6 - Linear Model Selection and Regularization

Problem #4

- (a) (iii) - Steadily increase.

Justification: As λ increases, the model's flexibility decreases due to the regularization term. This simplifies the model, but also makes it more biased. Consequently, the training RSS will steadily increase as λ becomes larger.

- (b) (ii) - Decrease initially, and then eventually start increasing in a U shape.

Justification: Test RSS will typically follow a similar trend to training RSS. It decreases initially as the model becomes less complex, but it might start increasing when λ becomes excessively large, causing underfitting.

- (c) (iv) - Steadily decrease.

Justification: Increasing λ reduces the model's complexity, leading to lower variance. The trend is for variance to steadily decrease as λ increases.

- (d) (iii) - Steadily increase.

Justification: Increasing λ encourages smaller coefficients, simplifying the model, and leading to an increase in bias, which steadily rises.

- (e) (v) - Remain constant.

Justification: The irreducible error represents the inherent noise in the data, which remains constant and is not affected by changes in the model or regularization parameter λ .

Problem #5

- (a) Ridge Regression Optimization Problem: The ridge regression aims to minimize the following objective function:

$$\min_{\beta} \sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p \beta_j^2$$

However, in this setting, the estimate for the intercept is zero, and the linear relationships are as follows: $y_1 + y_2 = 0$ and $x_{11} + x_{21} = 0$ and $x_{12} + x_{22} = 0$. So, $\hat{\beta}_0 = 0$, and the ridge regression optimization problem can be written as:

$$\min_{\beta} \sum_{i=1}^n \left(y_i - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \quad (1)$$

$$= \min_{\beta} \sum_{i=1}^2 \left(y_i - \sum_{j=1}^2 \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^2 \beta_j^2 \quad (2)$$

$$= \min_{\beta_1, \beta_2} (y_1 - \beta_1 x_{11} - \beta_2 x_{12})^2 + (y_2 - \beta_1 x_{21} - \beta_2 x_{22})^2 + \lambda (\beta_1^2 + \beta_2^2) \quad (3)$$

(b) To find the minimum of the ridge regression objective, we can take partial derivatives with respect to β_1 and β_2 . For this problem, we will set $x_1 = x_{11} = x_{12}$ and $x_2 = x_{21} = x_{22}$. After expanding the equation and taking the partial derivative, we have:

$$\frac{\partial}{\partial \beta_1} = -2y_1x_1 + 2\beta_1x_1^2 + 2\beta_2x_1^2 - 2y_2x_2 + 2\beta_1x_2^2 + 2\beta_2x_2^2 + 2\lambda\beta_1$$

$$\frac{\partial}{\partial \beta_2} = -2y_1x_1 + 2\beta_1x_1^2 + 2\beta_2x_1^2 - 2y_2x_2 + 2\beta_1x_2^2 + 2\beta_2x_2^2 + 2\lambda\beta_2$$

A minimum can be found when $\frac{\partial}{\partial \beta_1}$ and $\frac{\partial}{\partial \beta_2}$ are set to 0:

$$0 = 2y_1x_1 + 2y_2x_2 - 2\beta_1x_1^2 - 2\beta_2x_1^2 - 2\beta_1x_2^2 - 2\beta_2x_2^2 - 2\lambda\beta_1$$

$$0 = 2y_1x_1 + y_2x_2 - 2\beta_1x_1^2 - 2\beta_2x_1^2 - 2\beta_1x_2^2 - 2\beta_2x_2^2 - 2\lambda\beta_2$$

Divide both sides by 2λ ,

$$\beta_1 = (y_1x_1 + y_2x_2 - \beta_1x_1^2 - \beta_2x_1^2 - \beta_1x_2^2 - \beta_2x_2^2)/\lambda$$

$$\beta_2 = (y_1x_1 + y_2x_2 - \beta_1x_1^2 - \beta_2x_1^2 - \beta_1x_2^2 - \beta_2x_2^2)/\lambda$$

Therefore, in this setting, the ridge coefficient estimates satisfy $\beta_1 = \beta_2$. Ridge regression tends to give similar coefficient values to correlated variables due to the penalty term which pulls them towards each other.

(c) Lasso Optimization Problem: The lasso aims to minimize the following objective function:

$$\min_{\beta} \sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p |\beta_j|$$

In this setting with $\hat{\beta}_0 = 0$ and the given linear relationships, the lasso optimization problem can be written as:

$$\min_{\beta_1, \beta_2} \sum_{i=1}^n (y_i - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda(|\beta_1| + |\beta_2|)$$

(d) In this setting, the lasso coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ are not unique. There are many possible solutions to the optimization problem in (c). The lasso penalty term prefers sparsity, and in this case, there are multiple solutions where one or both of $\hat{\beta}_1$ and $\hat{\beta}_2$ can be equal to zero while still minimizing the objective function. The specific values of $\hat{\beta}_1$ and $\hat{\beta}_2$ may depend on the choice of λ and other factors, leading to different combinations of zero and non-zero coefficients. This characteristic of the lasso makes it suitable for variable selection and can lead to quite different coefficient values for correlated variables, unlike ridge regression.

Problem #7

(a) To derive the likelihood of the data, we are given the following model:

$$y_i = \beta_0 + \sum_{j=1}^p x_{ij}\beta_j + \epsilon_i$$

where ϵ_i follows a normal distribution with mean 0 and variance σ^2 . In mathematical terms:

$$\epsilon_i \sim N(0, \sigma^2)$$

Since the ϵ_i terms are independent and identically distributed, we can write the likelihood function as the product of the individual likelihoods for each observation:

$$\mathcal{L}(\beta_0, \beta_1, \dots, \beta_p, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \beta_0 - \sum_{j=1}^p x_{ij}\beta_j)^2}{2\sigma^2}\right)$$

(b) For the prior on β , we are given that $\beta_1, \beta_2, \dots, \beta_p$ are independent and identically distributed according to a double-exponential (Laplace) distribution with mean 0 and scale parameter b :

$$p(\beta) = \frac{1}{2b} \exp\left(-\frac{|\beta|}{b}\right)$$

The posterior for β is obtained using Bayes' theorem:

$$p(\beta|y, X) \propto \mathcal{L}(\beta|y, X) \cdot p(\beta)$$

The proportionality sign indicates that we are ignoring the normalizing constant. After plugging in the likelihood and prior, we have:

$$p(\beta|y, X) \propto \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \beta_0 - \sum_{j=1}^p x_{ij}\beta_j)^2}{2\sigma^2}\right) \cdot \prod_{j=1}^p \frac{1}{2b} \exp\left(-\frac{|\beta_j|}{b}\right)$$

(c) To show that the Lasso estimate is the mode for β under the posterior distribution, we can consider a Bayesian framework where we impose a Laplace (L1) prior on the coefficients β . In this context, the posterior distribution for β , given the data and the prior, is proportional to the likelihood and the prior. Specifically, the posterior distribution is given by:

$$P(\beta|Y, X) \propto P(Y|X, \beta) \cdot P(\beta)$$

Where:

- $P(\beta|Y, X)$ is the posterior distribution of the coefficients β given the data Y and the predictor variables X .
- $P(Y|X, \beta)$ is the likelihood of the data given the predictors and coefficients.
- $P(\beta)$ is the prior distribution of the coefficients.

In the context of the Lasso, we use a Laplace prior on the coefficients:

$$P(\beta) \propto \exp\left(-\lambda \sum_{i=1}^p |\beta_i|\right)$$

Where λ is the regularization parameter, and p is the number of predictors. The likelihood $P(Y|X, \beta)$ typically follows a Gaussian distribution:

$$P(Y|X, \beta) \propto \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta^T x_i)^2 \right)$$

Where σ is the standard deviation of the errors, n is the number of data points, and y_i and x_i are the observed data and predictor variables, respectively.

Now, to find the mode of the posterior distribution, we seek the values of β that maximize $P(\beta|Y, X)$, which can be done by maximizing the log of the posterior distribution, i.e., taking the logarithm of the right-hand side of the equation. Since taking the logarithm is a monotonic transformation, the mode remains the same:

$$\log P(\beta|Y, X) \propto -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta^T x_i)^2 - \lambda \sum_{i=1}^p |\beta_i|$$

To find the mode, we differentiate the log posterior distribution with respect to β and set the result equal to zero. The resulting optimization problem seeks to maximize the likelihood while simultaneously minimizing the L1 penalty due to the Laplace prior. This is precisely the objective function of the Lasso regression, which aims to find the mode (maximum a posteriori estimate) of β under this posterior distribution.

(d) For the prior on β , we are given that $\beta_1, \beta_2, \dots, \beta_p$ are independent and identically distributed according to a normal distribution with mean 0 and variance c :

$$p(\beta) = \prod_{j=1}^p \frac{1}{\sqrt{2\pi c}} \exp \left(-\frac{\beta_j^2}{2c} \right)$$

The posterior for β is still obtained using Bayes' theorem, as in part (b). After plugging in the likelihood and prior, we have:

$$p(\beta|y, X) \propto \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(y_i - \beta_0 - \sum_{j=1}^p x_{ij}\beta_j)^2}{2\sigma^2} \right) \cdot \prod_{j=1}^p \frac{1}{\sqrt{2\pi c}} \exp \left(-\frac{\beta_j^2}{2c} \right)$$

(e) To show that the Ridge Regression estimate is both the mode and the mean for β under the posterior distribution, we need to consider a Bayesian framework where we impose a Gaussian (L2) prior on the coefficients β . In this context, the posterior distribution for β , given the data and the prior, is proportional to the likelihood and the prior. Specifically, the posterior distribution is given by:

$$P(\beta|Y, X) \propto P(Y|X, \beta) \cdot P(\beta)$$

Where:

- $P(\beta|Y, X)$ is the posterior distribution of the coefficients β given the data Y and the predictor variables X .

- $P(Y|X, \beta)$ is the likelihood of the data given the predictors and coefficients.
- $P(\beta)$ is the prior distribution of the coefficients.

In the context of Ridge Regression, we use a Gaussian prior on the coefficients:

$$P(\beta) \propto \exp \left(-\frac{1}{2\tau^2} \sum_{i=1}^p \beta_i^2 \right)$$

Where τ is the regularization parameter, and p is the number of predictors.

The likelihood $P(Y|X, \beta)$ typically follows a Gaussian distribution:

$$P(Y|X, \beta) \propto \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta^T x_i)^2 \right)$$

Where σ is the standard deviation of the errors, n is the number of data points, and y_i and x_i are the observed data and predictor variables, respectively.

Now, to find the mode of the posterior distribution, we differentiate the log of the posterior distribution with respect to β and set the result equal to zero. This optimization problem seeks to maximize the likelihood while minimizing the L2 penalty due to the Gaussian prior. The solution is the Ridge Regression estimate for β .

Additionally, when a Gaussian prior is used, the Ridge Regression estimate is also the mean of the posterior distribution. This is a characteristic of the Gaussian distribution and the property that the mean (expectation) of a Gaussian distribution is the maximum of its probability density function.

In summary, Ridge Regression provides the mode (maximum a posteriori estimate) and the mean for β under the posterior distribution, which is characterized by a Gaussian prior on the coefficients. This is a unique property of Ridge Regression within the Bayesian framework.

Problem #9

(a) Done

(b) the MSE is approximately 1040120.0912491741, which means, on average, the squared difference between the predicted and true values is around 1,040,120. Also, the linear regression model explains approximately 89.85% of the variance in the test data. This is a relatively high R^2 score and suggests that the model is doing a good job of explaining the variability in the target variable.

(c) The squared difference between the predicted values and the true values is around 1,009,951 for the scaled test data. This MSE is similar in magnitude to the previous example, indicating that the rcv model performs well, as it has a relatively low mean squared error. Also, the rcv model explains approximately 90.15% of the variance in the scaled test data. This is a relatively high R^2 score and suggests that the rcv model is performing well in explaining the variability in the target variable for

the scaled data.

(d) On average, the squared difference between the predicted values and the true values is around 1,023,952 for the scaled test data. The lcv model explains approximately 90.01% of the variance in the scaled test data. This is a relatively high R^2 score, similar to the previous examples, suggesting that the lcv model is doing a good job of explaining the variability in the target variable for the scaled test data. (g) To evaluate the results and the accuracy of predicting the number of college applications received, we need to consider the test errors obtained from the five different approaches you've shown. Let's summarize and compare the results:

1. **lr** Model:

- Mean Squared Error (MSE): Approximately 1,040,120.0912
- R^2 Score: 0.8985

2. **rcv** Model:

- Mean Squared Error (MSE): Approximately 1,009,951.4008
- R^2 Score: 0.9015

3. **lcv** Model:

- Mean Squared Error (MSE): Approximately 1,023,952.7020
- R^2 Score: 0.9001

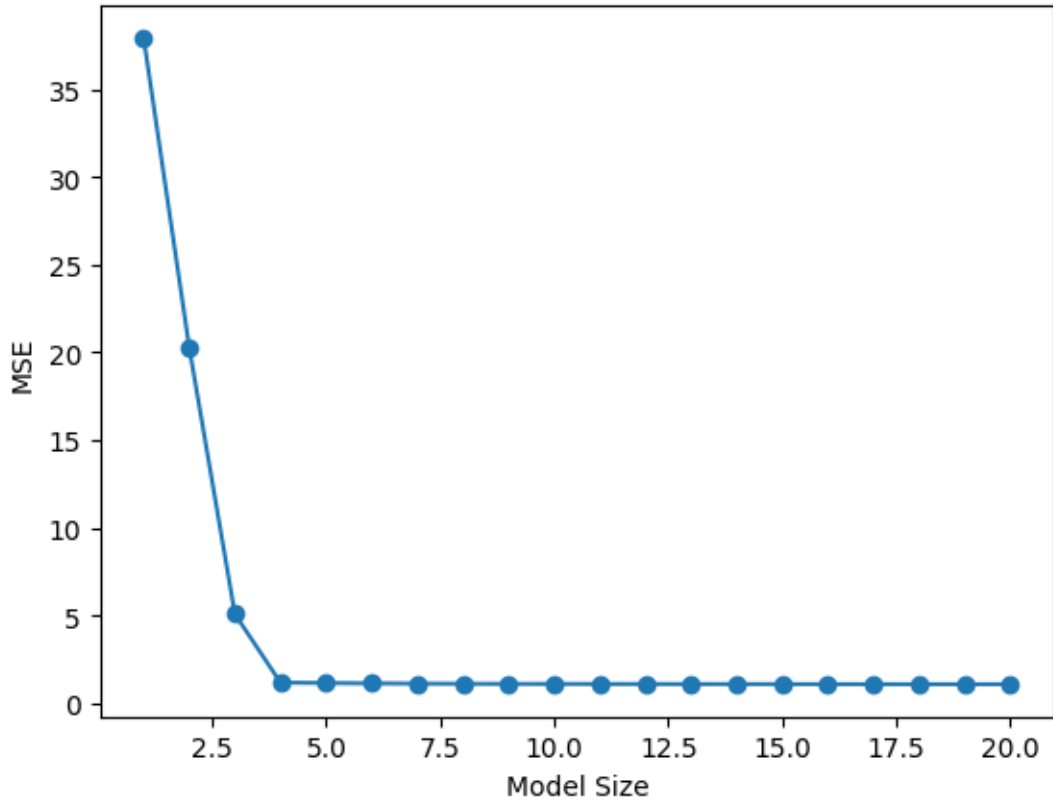
These results suggest that all three models (linear regression, RidgeCV, and LassoCV) are performing quite well in predicting the number of college applications received. The R^2 scores are all around 0.90, which indicates that these models explain approximately 90% of the variance in the test data, and the mean squared errors are all in the range of approximately 1,000,000. These errors are relatively low considering the magnitude of the data.

Comparing the models, the RidgeCV model (**rcv**) has the lowest mean squared error (MSE), indicating slightly better predictive performance in terms of reducing squared prediction errors. However, the differences in MSE between the models are not substantial.

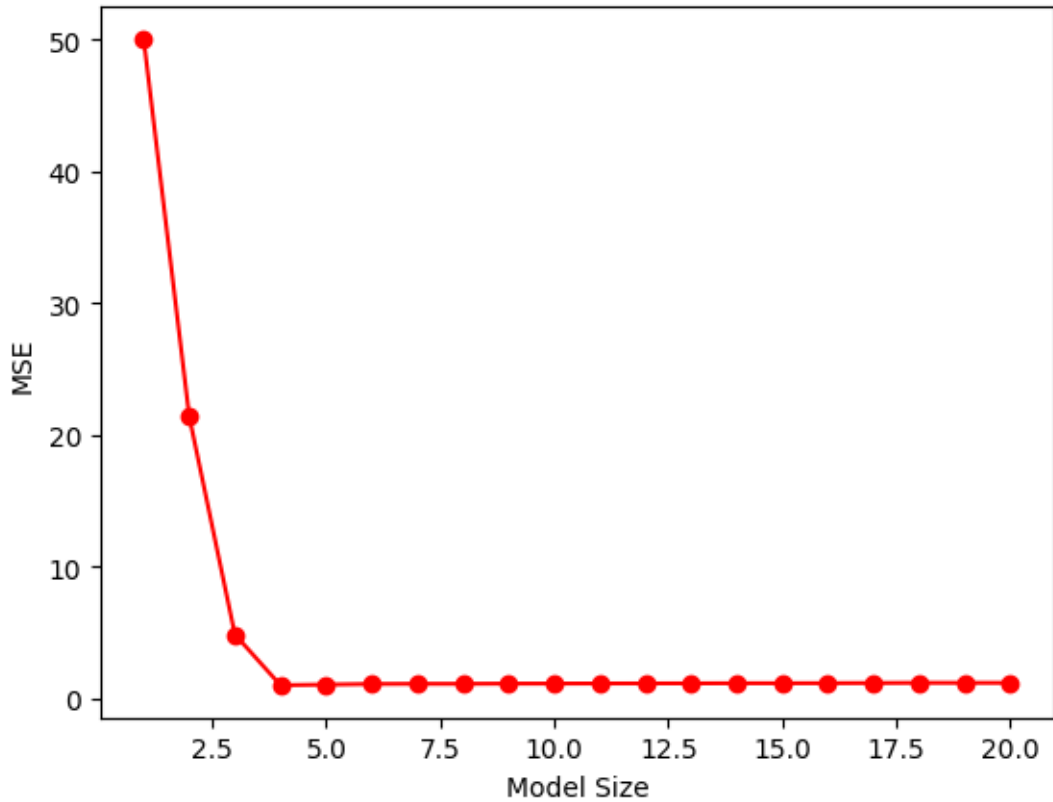
In summary, all five approaches perform quite well, with a high degree of accuracy in predicting the number of college applications received.

Problem #11

- (a) Done
- (b) Done
- (c) Training MSE:



(d) Test MSE:

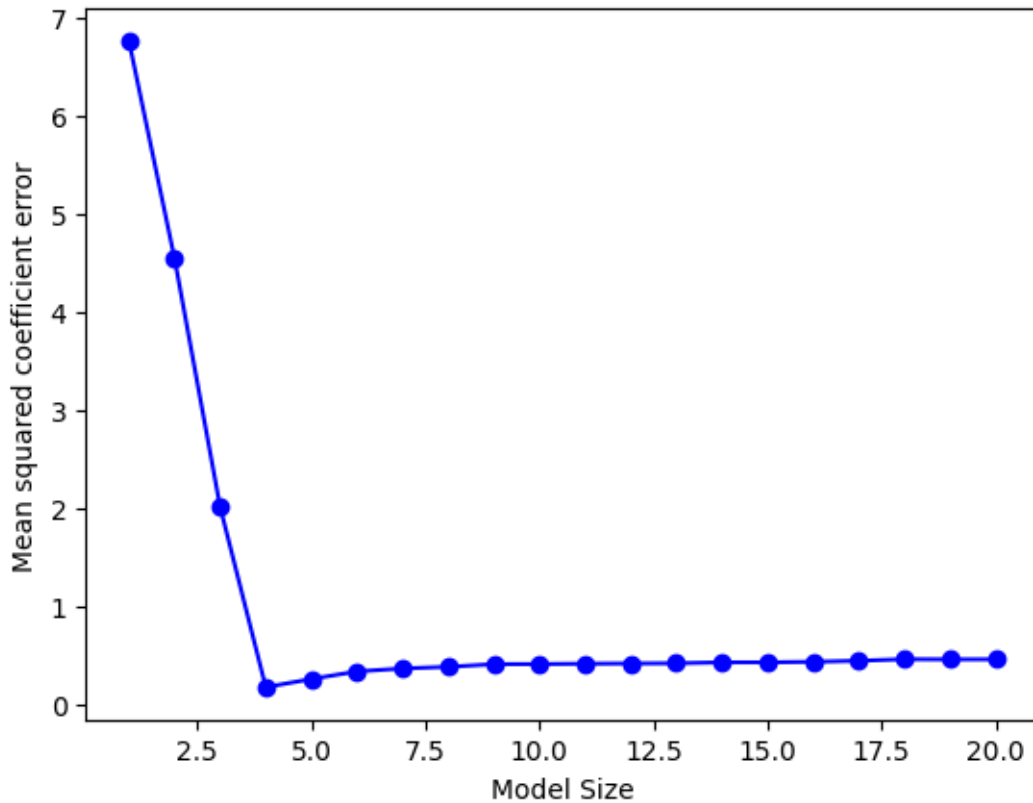


(e) The minimum test Mean Squared Error (MSE) is observed for a model of size 4. This corresponds to the four non-zero coefficients we originally defined in the true model $\beta_{0:4} = [5, 4, 2, 7]$. Hence, the best subset selection method correctly identified the number of significant predictors.

(f) The estimated coefficients are 4.93, 4.01, 1.85, and 7.04. Comparing these with the true coefficients

[5, 4, 2, 7], we see that the estimates are quite close. This is a good indication of the model's accuracy. Small discrepancies are expected due to the noise (ϵ) introduced in the data.

(g) MSCE:



The optimal model size, in terms of minimizing test MSE, is 4. This conclusion is drawn from the second graph (red) which displays the test MSE. The MSE sharply decreases as the model size increases, reaching its lowest point at model size 4, after which it remains relatively stable. Similarly, the training MSE (first graph in blue) and MSCE (third graph in blue) also show a sharp decline up to model size 4, suggesting that this model size offers a good trade-off between fitting the training data and generalizing to new data while providing accurate coefficient estimates. Increasing the model size beyond 4 does not yield significant reductions in any of the metrics, indicating that additional complexity might not be beneficial.

Chapter 7 - Moving Beyond Linearity

Problem #1

(a) We want $f_1(x)$ to be equal to $f(x)$ for all $x \leq \xi$. This means that for $x \leq \xi$, the term $\beta_4(x - \xi)_+^3$ becomes zero. So, for $x \leq \xi$, we have:

$$f_1(x) = \beta_0 + \beta_1x + \beta_2x^2 + \beta_3x^3$$

This is a cubic polynomial in the form $f_1(x) = a_1 + b_1x + c_1x^2 + d_1x^3$, where:

- $a_1 = \beta_0$
- $b_1 = \beta_1$
- $c_1 = \beta_2$
- $d_1 = \beta_3$

(b) For $x > \xi$, we have:

$$f_2(x) = \beta_0 + \beta_1x + \beta_2x^2 + \beta_3x^3 + \beta_4(x - \xi)^3$$

For this part, we need to isolate the terms that are not in $f_1(x)$. These are the terms involving $\beta_4(x - \xi)^3$:

$$f_2(x) = \beta_0 + \beta_1x + \beta_2x^2 + \beta_3x^3 + \beta_4(x - \xi)^3$$

So, for $x > \xi$, we have:

$$\begin{aligned} f_2(x) &= \beta_0 + \beta_1x + \beta_2x^2 + \beta_3x^3 + \beta_4(x^3 - 3x^2\xi + 3x\xi^2 - \xi^3) \\ f_2(x) &= (\beta_0 - \beta_4\xi^3) + (\beta_1 + 3\beta_4\xi^2)x + (\beta_2 - 3\beta_4\xi)x^2 + (\beta_3 + \beta_4)x^3 \end{aligned}$$

This is still a cubic polynomial in the form $f_2(x) = a_2 + b_2x + c_2x^2 + d_2x^3$, where:

- $a_2 = \beta_0 - \beta_4\xi^3$
- $b_2 = \beta_1 + 3\beta_4\xi^2$
- $c_2 = \beta_2 - 3\beta_4\xi$
- $d_2 = \beta_3 + \beta_4$

(c)

$$f_1(\xi) = \beta_0 + \beta_1\xi + \beta_2\xi^2 + \beta_3\xi^3$$

$$f_2(\xi) = (\beta_0 - \beta_4\xi^3) + (\beta_1 + 3\beta_4\xi^2)\xi + (\beta_2 - 3\beta_4\xi)\xi^2 + (\beta_3 + \beta_4)\xi^3$$

Now, let's simplify and compare these expressions:

$$f_1(\xi) = \beta_0 + \beta_1\xi + \beta_2\xi^2 + \beta_3\xi^3$$

$$f_2(\xi) = (\beta_0 - \beta_4\xi^3) + (\beta_1 + 3\beta_4\xi^2)\xi + (\beta_2 - 3\beta_4\xi)\xi^2 + (\beta_3 + \beta_4)\xi^3$$

Now, notice that all terms have a common factor of ξ^3 , which cancels out:

$$f_1(\xi) = \beta_0 + \beta_1\xi + \beta_2\xi^2 + \beta_3\xi^3$$

$$f_2(\xi) = \beta_0 - \beta_4\xi^3 + \beta_1\xi + \beta_2\xi^2 + \beta_3\xi^3 + \beta_4\xi^3$$

So, after canceling the common factor, we have:

$$f_1(\xi) = \beta_0 + \beta_1\xi + \beta_2\xi^2 + \beta_3\xi^3 = f_2(\xi)$$

Therefore, $f(x)$ is continuous at ξ .

(d) First, compute the derivatives:

$$f'_1(x) = b_1 + 2c_1x + 3d_1x^2$$

$$f'_2(x) = b_2 + 2c_2x + 3d_2x^2$$

Now, evaluate these derivatives at ξ :

$$f'_1(\xi) = b_1 + 2c_1\xi + 3d_1\xi^2$$

$$f'_2(\xi) = b_2 + 2c_2\xi + 3d_2\xi^2$$

From the corrected expressions in (a) and (b), we know the values of b_1 , c_1 , d_1 , b_2 , c_2 , and d_2 . Substitute these values into the above equations:

$$f'_1(\xi) = \beta_1 + 2\beta_2\xi + 3\beta_3\xi^2$$

$$f'_2(\xi) = (\beta_1 + 3\beta_4\xi^2) + 2(\beta_2 - 3\beta_4\xi)\xi + 3(\beta_3 + \beta_4)\xi^2$$

Now, let's simplify these expressions:

$$f'_1(\xi) = \beta_1 + 2\beta_2\xi + 3\beta_3\xi^2$$

$$f'_2(\xi) = \beta_1 + 2\beta_2\xi + 3\beta_3\xi^2$$

As you can see, $f'_1(\xi) = f'_2(\xi)$, so the first derivative of $f(x)$ is continuous at ξ .

(e) Compute the second derivatives:

$$f_1''(x) = 2c_1 + 6d_1x$$

$$f_2''(x) = 2c_2 + 6d_2x$$

We know the values of c_1 , c_2 , d_1 , and d_2 . Substitute these values into the above equations:

$$f_1''(\xi) = 2\beta_2 + 6\beta_3\xi$$

$$\begin{aligned} f_2''(\xi) &= \beta_1 + 3\beta_4\xi^2 + 2(\beta_2 - 3\beta_4\xi)\xi + 3(\beta_3 + \beta_4)\xi^2 \\ &= \beta_1 + 3\beta_4\xi^2 + 2\beta_2\xi - 6\beta_4\xi^2 + 3\beta_3\xi^2 + 3\beta_4\xi^2 \\ &= \beta_1 + 2\beta_2\xi + 3\beta_3\xi^2 \end{aligned}$$

As we can see, $f_1''(\xi) = f_2''(\xi)$, so the second derivative of $f(x)$ is continuous at ξ .

Therefore, we have shown that $f(x)$ is indeed a cubic regression spline, as it is piecewise continuous and has continuous first and second derivatives at ξ .

Problem #2

(a) $\lambda = \infty$, $m = 0$:

When λ is set to infinity, it means that the regularization term has an extremely strong influence on the curve. In this case, the function will be always evaluated to 0.

(b) $\lambda = \infty$, $m = 1$:

With λ set to infinity and $m = 1$, for $\hat{g}$, the slope will be 0. So it will be a constant function.

(c) $\lambda = \infty$, $m = 2$:

For $m = 2$, $\hat{g}$ will attempt to be a straight line.

(d) $\lambda = \infty$, $m = 3$:

With $m = 3$ and λ set to infinity, $\hat{g}$ will be a quadratic (second-degree polynomial) function. The curve will be even smoother than in the previous cases, and it will try to minimize any variations in the curvature.

(e) $\lambda = 0$, $m = 3$:

When λ is set to 0, there is no regularization, and the curve will try to fit the data points as closely as possible, regardless of smoothness. In this case, with $m = 3$, $\hat{g}$ will be a cubic polynomial that closely interpolates the data points without considering smoothness, leading to a curve that will pass through each data point exactly.

In all these scenarios, the value of λ and the order of the derivative m play a crucial role in determining how the curve $\hat{g}$ behaves. Higher values of λ emphasize smoothness, while lower values focus on fitting the data closely. The choice of m determines the degree of the polynomial that $\hat{g}$ will approximate.

Problem #4

$$\text{For } X < 0 : \quad \hat{Y} = 1$$

$$\text{For } 0 \leq X < 1 : \quad \hat{Y} = 2$$

$$\text{For } 1 \leq X \leq 2 : \quad \hat{Y} = 3 - X$$

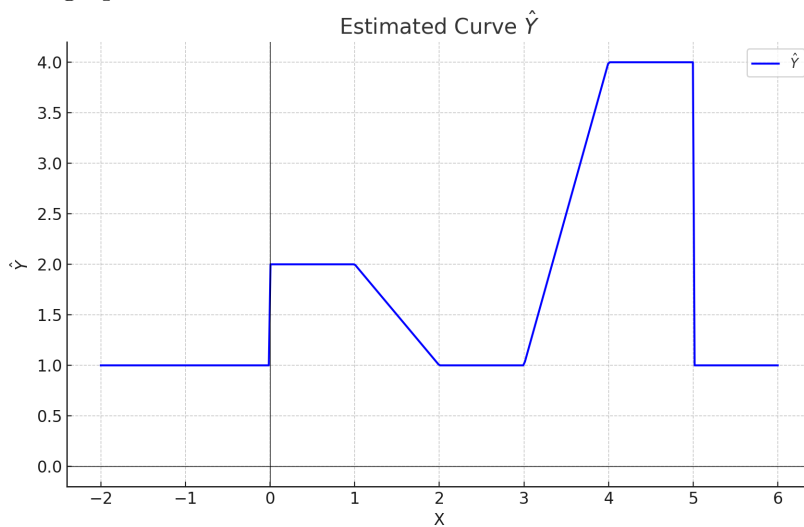
$$\text{For } 2 < X < 3 : \quad \hat{Y} = 1$$

$$\text{For } 3 \leq X \leq 4 : \quad \hat{Y} = 3X - 8$$

$$\text{For } 4 < X \leq 5 : \quad \hat{Y} = 4$$

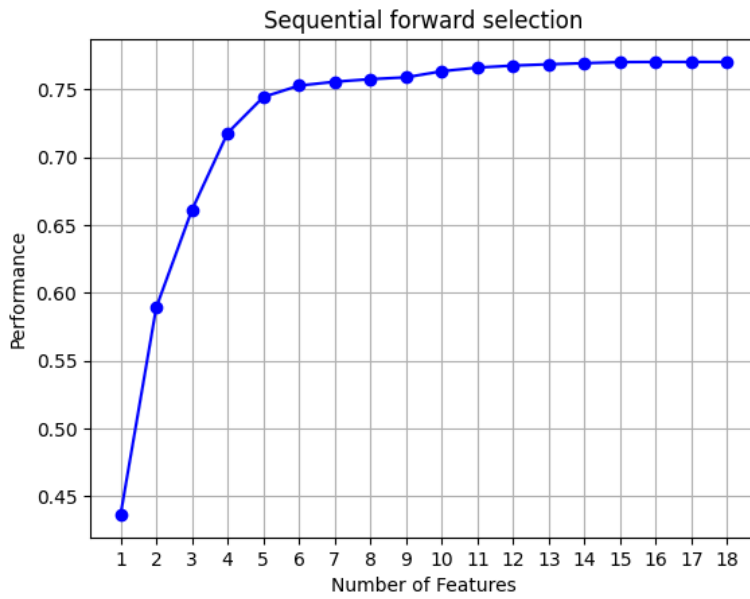
$$\text{For } X > 5 : \quad \hat{Y} = 1$$

The graph:



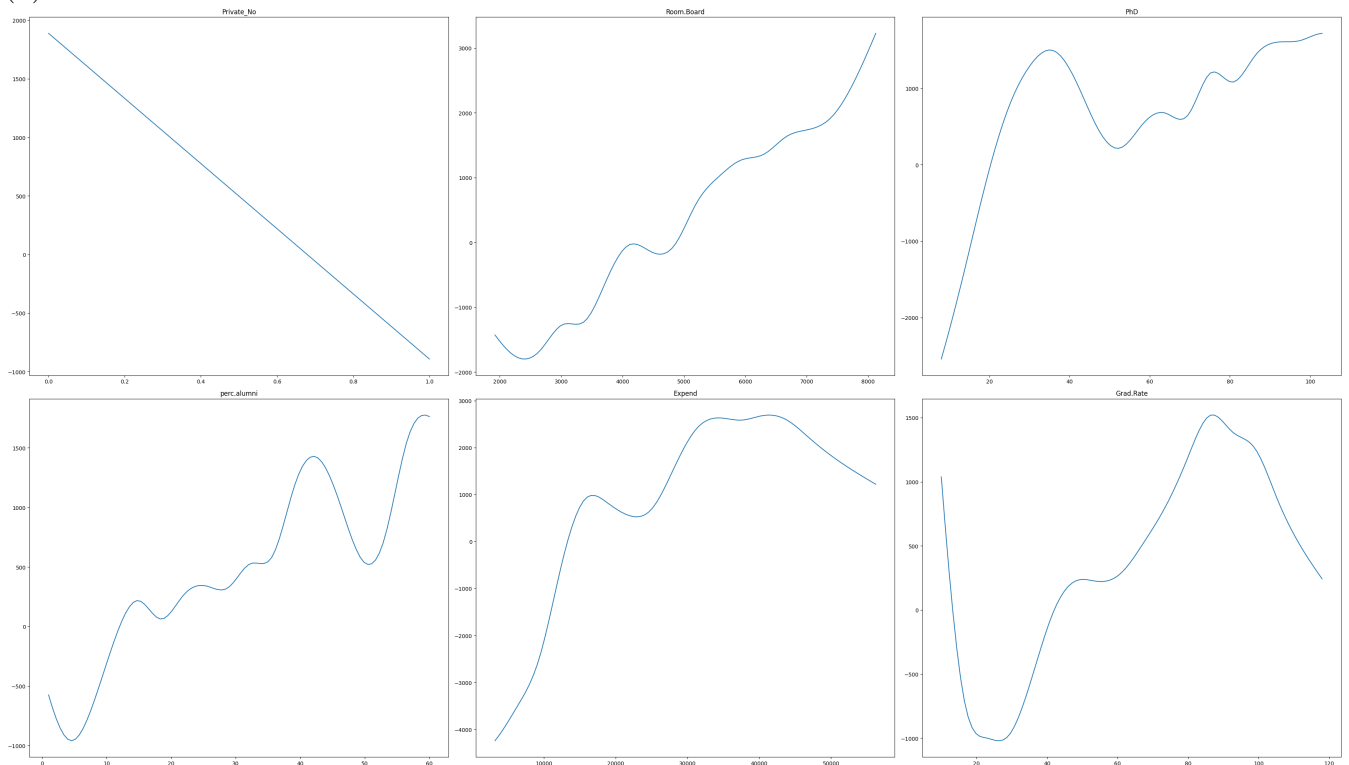
Problem #10

(a)



From the graph, we estimate 6 to be enough features. The best variables for these are Private_No, Room.Board, PhD, perc alumni, Expend, Grad.Rate.

(b)



(c)

MAE: 1458.6163982208898

it states that the average error in terms of absolute value is \$1458 between the predicted tuition fee and the actual tuition fee.

MSE: 3481100.4081448447

RMSE: 1865.7707276471149

RMSE (which is a better indicator than MSE), usually also works better than just raw MAE. Here we have approximately \$1865 as RMSE. It implies an average of \$1865 error in predicting the tuition cost. R-squared (R2): 0.7896781388844216

As R-squared value is 0.7896, this model can capture 78.96% of the variance in the dataset.

(d) From gam summary, Expend, Room.Board and perc.alumni seems to have the highest non-linearity.

Problem #11

(a) Done

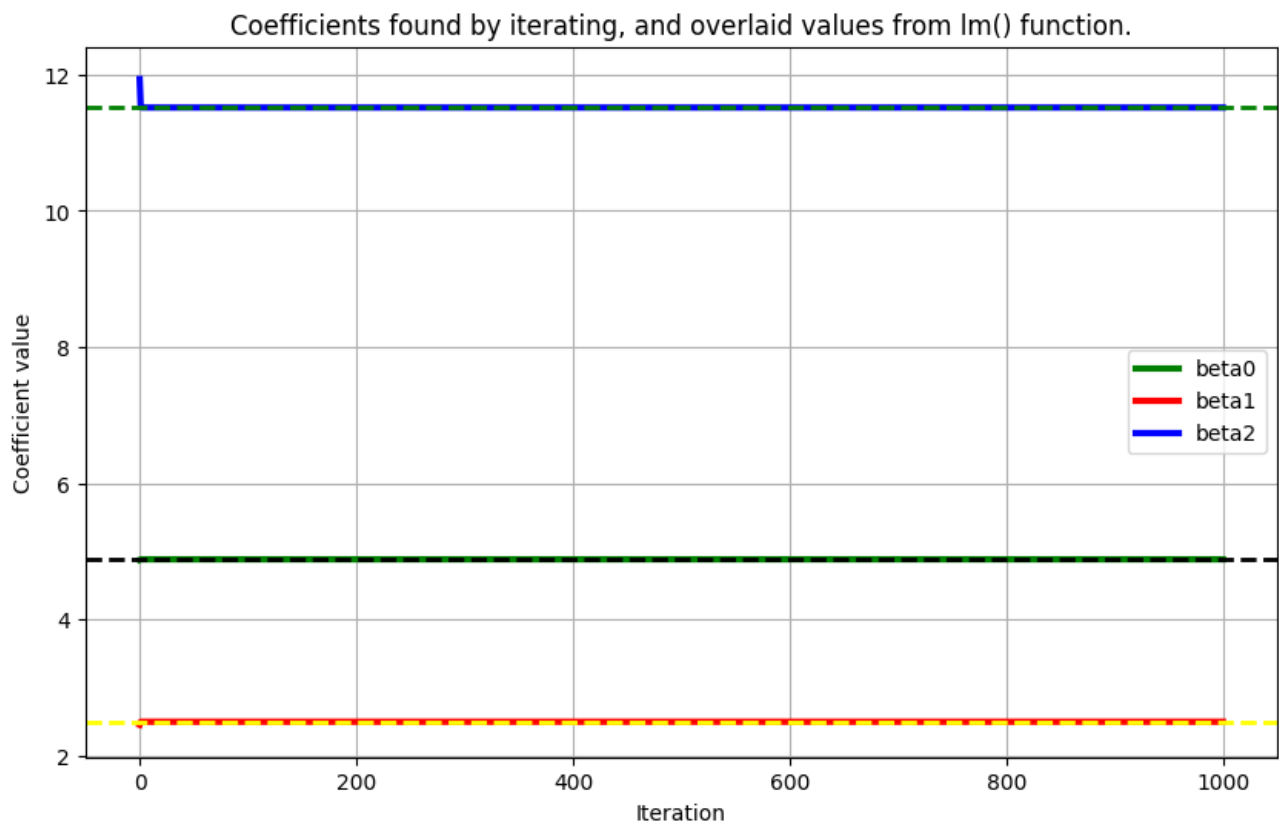
(b) Done

(c) Done

(d) Done

(e) Done

(f, g)



(h) The green solid line represents beta0. It converges to a value slightly below 5 and stays consistent for most of the iterations.

The red solid line represents β_1 . It starts near 0, rapidly increases, and converges to a value slightly above 2.5.

The blue solid line represents β_2 . It quickly converges to a value slightly above 11.5 and remains stable throughout.

The dashed lines represent the coefficients derived from direct multiple linear regression using the lm() function. They serve as a benchmark.

Observations:

The coefficients converge relatively quickly. This is evident as the lines stabilize early on, and for most iterations, they remain consistent.

The coefficients derived from the iterative back-fitting approach closely match those derived from direct multiple linear regression.

This demonstrates that the back-fitting method can indeed approximate multiple linear regression when only simple linear regression tools are available.

By around the 5th iteration, the coefficients seem to have achieved a good approximation to the multiple linear regression results, which is efficient.